The association between vaping and subsequent initiation of cigarette smoking in young Australians from age 12 to 17 years: a retrospective cohort analysis using cross-sectional recall data from 5114 adolescents

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Abstract

Objective: The objective of this study was to estimate the association between vaping and subsequent initiation of smoking among Australian adolescents and explore the impact of design and analytical methods in previous studies.

Methods: We conducted a retrospective cohort analysis of cross-sectional data from 5114 Australian adolescents aged 14–17 recalling information on smoking and vaping initiation from age 12 to 17. The outcome was smoking initiation, analysed with negative-binomial regression to estimate incidence rate ratios (IRRs) for vape status (ever-vaped vs never-vaped) as a time-varying exposure. We also re-analysed using the methods of previous studies not accounting for the time-varying nature of e-cigarette exposure.

Results: Participants (n=5114) were retrospectively followed for 20478 person-years. After adjusting for socio-demographic variables and proxy measures of common liabilities for vaping and smoking, the rate of smoking initiation for those who ever-vaped was nearly 5 times that of those who never-vaped (IRR=4.9; 95% confidence interval: [3.9, 6.0], $p<0.001$), with IRRs considerably higher at younger ages. Not accounting for the time-varying nature of e-cigarette exposure in re-analysis attenuated the estimated IRR by 44%.

Conclusions: Controlled analyses indicate that vaping markedly increases the risk of subsequent smoking initiation among Australian adolescents from age 12 to 17, with those aged 12, 13, and 14 bearing an alarmingly disproportionate burden of the elevated risk. Additionally, the relative risk of future smoking due to vaping may have been underestimated in other studies due to methodological differences.

Implications for public health: Our findings highlight the need for public health interventions and strict e-cigarette access laws.

Key words: e-cigarettes, smoking, vaping, adolescents, Australian

Introduction

In recent decades, Australia has successfully reduced smoking prevalence via comprehensive tobacco control measures.[1] From 1980 to 2019, for example, the adult regular smoking prevalence declined from 35% to 13%.[2] Because most smokers start during their teenage years,[3] reducing smoking uptake among adolescents has been pivotal for Australia's success. In 1996, 58% of Australian 12- to 17-year-old students had tried smoking; by 2022/23, this had dropped to 14%.[4] The 2012 US Surgeon General's report noted that "preventing smoking… among young people is critical to ending the epidemic of tobacco use".[5] This underscores the importance of preventing rises in smoking uptake among Australian adolescents for Australia to continue its successful trajectory in tobacco control.

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Contrasting Australia's tobacco control success, population surveillance data suggest rapidly increasing rates of e-cigarette use (vaping), particularly among teenagers and young adults.[6,7] In 2014, for example, the Australian Secondary Students' Alcohol and Drug (ASSAD) survey found that 13% of 12- to 17-year-old Australian secondary school students had tried an e-cigarette,[7] and by 2022/23, this had risen to 30%.[7] This dramatic rise has occurred notwithstanding 'conclusive evidence' that e-cigarettes can directly cause harm to human health through lung injury, poisoning, burns, seizures, and various other acute events.[8] In a recent survey of 4695 current and former e-cigarette users, for example, 78.9% of respondents reported having had at least one adverse event within 6 hours of use, including symptoms such as headache, anxiety, and coughing.[9]

In addition to the direct harms, there is also 'strong evidence' that vaping can indirectly harm adolescent users by influencing their future smoking behaviour. In the most recent systematic review on the topic,[10] a 2021 meta-analysis of 17 studies found that among people aged less than 20 years, ever-use of e-cigarettes was associated with a three-fold increased risk of subsequent smoking uptake. Although a three-fold increased risk is concerning, this estimate is potentially conservative because, as far as we are aware, only one study on this topic accounted for the time-varying nature of e-cigarette exposure.[11] Not accounting for time-varying exposure variables risks underestimation of associations through estimate attenuation (or dilution),[12,13] particularly for exposure variables with high rates of within-individual change during the study follow-up period. A seemingly pertinent example of such an exposure variable is whether an adolescent has ever tried vaping.

Previously, we reported cross-sectional associations between ever-smoking and ever-vaping among 721 14- to 17-year-old participants of wave 1 of the *Generation Vape* study.[14] These findings, while informative, did not address the chronological order of smoking and vaping initiations, necessary for establishing cause and effect. To overcome this limitation, new questions were added to the questionnaire from wave 4 onwards, asking participants to recall age(s) at smoking/vaping initiations. These questions complemented an existing question asking participants to recall the order of smoking/vaping initiations, thereby allowing us to construct retrospective timelines of initiation events. These timelines are used here to undertake a retrospective cohort analysis of cross-sectional data[15] to assess whether vaping initiation is associated with subsequent smoking uptake among Australian adolescents. Notably, this relationship has not been previously explored in an Australian adolescent cohort (either retrospective or prospective), thus making it one of the aims of this paper. Additionally, we aim to assess if the design and analytical methods of previous studies investigating this relationship in other countries might have impacted their findings.

Methods

Study design

Data for the current analyses were drawn from the larger *Generation Vape* study, which examines vaping and smoking behaviours in four population groups (14- to 17-year-olds, 18- to 24-year-olds, parents of 14- to 17-year-olds and secondary school educators) at multiple 6-monthly timepoints (waves). Study participants are recruited from online panels of Australia-wide residents and are mostly unique across waves but with some repeat participants.

Participants

For the current analyses, we included 14- to 17-year-old respondents of the wave 4 and 5 surveys (conducted February–April and August–September 2023) following the introduction of the new questionnaire items mentioned earlier. If adolescents participated in both waves, only their later data were included. Parent and adolescents' consent was required for adolescents' participation.

Measures

Socio-demographic variables

Demographic data used in these analyses included gender, age, Aboriginal and/or Torres Strait Islander status, state/territory of residence, remoteness of residence (metro vs. non-metro areas of Australia), socioeconomic status (SES) of residence area (divided into quintiles),[16] and language spoken at home (English and/or another language) (Table 1).

Common liabilities for vaping and smoking

We derived four 'common liability' variables from survey responses to serve as potential confounders in the relationship between vaping and smoking. These variables reflect shared underlying risks or predispositions affecting both vaping and smoking behaviours. The first measured whether participants' friends smoke and/or vape, identifying those with no friends who smoke or vape, those with friends who only vape, those with friends who only smoke, and those whose friends do both. The second measured participants' perceptions of family/friends' views on smoking and vaping, ranging from not approving of either, to approval of one but not the other, and approval of both. The third reflects participants' awareness of the harms of nicotine, gauged by their agreement—measured on a 5-point Likert scale—with the statement "nicotine is harmless". The fourth reflects whether the participant is still at school, pursuing other vocational activities or has left school and not pursuing other vocational activities.

Time-at-risk of smoking initiation and vape status

Using responses to questions about age and order of smoking/vaping initiations, we constructed retrospective timelines of initiation events across single-year age intervals starting at age 12 for each participant (**Figure 1**). Timelines started at age 12, as only 13 participants began smoking earlier resulting in unreliable age-specific estimates before age 12. Smoking initiation was defined in the survey as having ever consumed at least "a few puffs" of "cigarettes, roll your own (rollies), cigars, shisha [or] pipes", and vaping initiation was defined as having ever consumed at least "a few puffs" of "vapes" or "e-cigarettes". Participants were retrospectively followed from the start of age-interval 12 to midway through the 'end-of-follow-up' age interval, with end-of-follow-up occurring at smoking initiation, or survey completion for non-initiators. The variable 'time-at-risk' of smoking initiation was calculated according to the primary exposure variable 'vape status', a time-varying covariate defined as 'never-vaped' or 'ever-vaped' for periods before and after (any) vaping initiation, respectively. For each participant, time-at-risk was set to 1 person-year for each single-year age interval prior to the end-of-follow-up age

Table 1: Characteristics of study participants in wave 4/5 (2023).

Characteristic	Ever-vaped prior to end of follow-up[a]/total participants	Initiated smoking/total participants	Characteristic distributions n (column %)
Total:	**1251/5114 (24%)**	**577/5114 (11%)**	**5114 (100%)**
Gender			
Male	480/2353 (20%)	226/2353 (10%)	2353 (46%)
Female	749/2662 (28%)	336/2662 (13%)	2662 (52%)
Non-binary and other	22/99 (22%)	15/99 (15%)	99 (2%)
Age at time of completing questionnaire (years)			
14	140/940 (15%)	36/940 (4%)	940 (18%)
15	245/1245 (20%)	96/1245 (8%)	1245 (24%)
16	378/1405 (27%)	178/1405 (13%)	1405 (27%)
17	488/1524 (32%)	267/1524 (18%)	1524 (30%)
Wave			
4 (Feb–Apr 2023)	551/2178 (25%)	291/2178 (13%)	2178 (43%)
5 (Aug–Sep 2023)	700/2936 (24%)	286/2936 (10%)	2936 (57%)
Aboriginal or Torres Strait Islander			
No	1194/4900 (24%)	522/4900 (11%)	4900 (96%)
Yes	50/184 (27%)	47/184 (26%)	184 (4%)
Prefer not to say	7/30 (23%)	8/30 (27%)	30 (1%)
State/territory of residence			
NSW	302/1253 (24%)	118/1253 (9%)	1253 (25%)
Vic	323/1346 (24%)	141/1346 (10%)	1346 (26%)
Qld	312/1180 (26%)	162/1180 (14%)	1180 (23%)
SA	117/446 (26%)	56/446 (13%)	446 (9%)
WA	154/686 (22%)	71/686 (10%)	686 (13%)
Tas/NT/ACT	43/203 (21%)	29/203 (14%)	203 (4%)
Remoteness of residence			
Metro	999/4094 (24%)	412/4094 (10%)	4094 (80%)
Non-metro	248/1008 (25%)	164/1008 (16%)	1008 (20%)
Not known	4/12 (33%)	1/12 (8%)	12 (0%)
SES of residence area			
1—lowest SES	165/638 (26%)	75/638 (12%)	638 (12%)
2	176/674 (26%)	103/674 (15%)	674 (13%)
3	247/996 (25%)	120/996 (12%)	996 (19%)
4	296/1235 (24%)	123/1235 (10%)	1235 (24%)
5—highest SES	363/1559 (23%)	155/1559 (10%)	1559 (30%)
Not known	4/12 (33%)	1/12 (8%)	12 (0%)
Language spoken at home			
English only	1101/4398 (25%)	529/4398 (12%)	4398 (86%)
Another language only	14/97 (14%)	10/97 (10%)	97 (2%)
English and another language	136/619 (22%)	38/619 (6%)	619 (12%)
Friends smoke or vape			
No friends vape or smoke	192/2500 (8%)	48/2500 (2%)	2500 (49%)
Friends vape but not smoke	672/1637 (41%)	173/1637 (11%)	1637 (32%)
Friends smoke but not vape	7/66 (11%)	8/66 (12%)	66 (1%)
Friends both vape and smoke	380/911 (42%)	348/911 (38%)	911 (18%)
Family/friends views on smoking/vaping			
Do not approve of smoking or vaping	1067/4623 (23%)	486/4623 (11%)	4623 (90%)
Approve of smoking but not vaping	63/165 (38%)	30/165 (18%)	165 (3%)
Approve of vaping but not smoking	86/219 (39%)	36/219 (16%)	219 (4%)
Approve of both smoking and vaping	35/107 (33%)	25/107 (23%)	107 (2%)
Agree with "nicotine is harmless"			
Strongly disagree	398/2312 (17%)	149/2312 (6%)	2312 (45%)
Disagree	504/1818 (28%)	246/1818 (14%)	1818 (36%)
Neither agree nor disagree	193/482 (40%)	108/482 (22%)	482 (9%)
Agree	100/287 (35%)	49/287 (17%)	287 (6%)
Strongly agree	56/215 (26%)	25/215 (12%)	215 (4%)

(continued)

Table 1. Continued

Characteristic	Ever-vaped prior to end of follow-up[a]/total participants	Initiated smoking/total participants	Characteristic distributions n (column %)
School/vocation commitment			
Still in school	1134/4797 (24%)	481/4797 (10%)	4797 (94%)
Other vocational pathway	65/156 (42%)	38/156 (24%)	156 (3%)
Not in school or other vocational pathway	52/161 (32%)	58/161 (36%)	161 (3%)

Abbreviations: NSW = New South Wales; NT = Northern Territory; WA = Western Australia; Qld = Queensland; Vic = Victoria; Tas = Tasmania; SA = South Australia; ACT = Australian Capital Territory; SES = socioeconomic status.

[a]An additional 243 adolescents initiated vaping after smoking initiation but are not counted as 'Ever-vaped prior to end of follow-up' because vaping initiation occurred after end-of-follow-up (see Figure 2, Person 6 for illustration). Hence, the proportion of adolescents who initiated vaping by time of survey was 29% (= [243 + 1251]/5114), higher than the 24% who initiated vaping during follow-up.

interval. For the end-of-follow-up age interval, the time-at-risk was set to 1/2 person-year because end of follow-up is assumed to have occurred, on average, halfway through the age interval. For age intervals where vaping was initiated prior to end of follow-up, time at risk was divided equally between never-vaped and ever-vaped categories.

Statistical methods

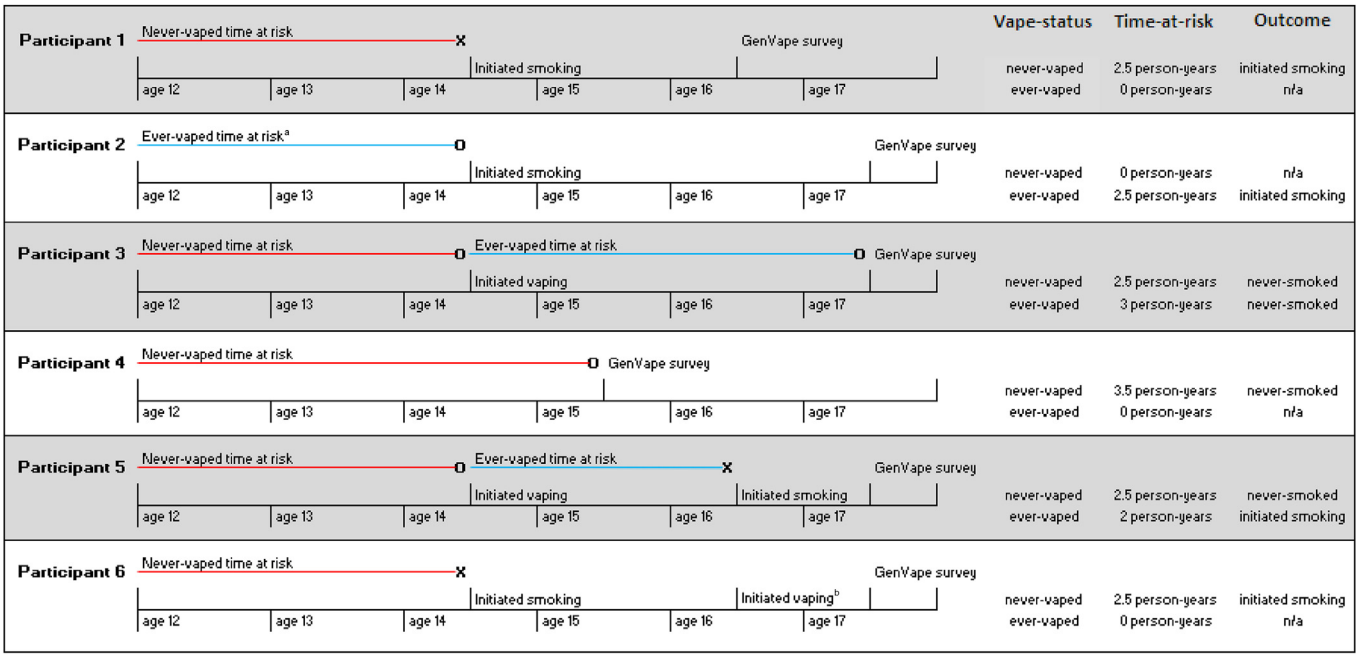
Main analysis

For each participant, data were split into rows for each single-year age interval between 12 and end of follow-up, with further splitting for age intervals if vaping was initiated (to allocate time-at-risk within the age interval equally between never-vaped and ever-vaped categories). Negative-binomial regression models were then used to assess the association between vape status (as a time-varying covariate) and subsequent smoking initiation (outcome variable), with time-at-risk of smoking initiation entered as the time variable and with incidence rate ratio (IRR) as the effect measure. Modelling vape status as a time-varying covariate accounts for its time-varying nature in the analysis.[17] Cluster robust variance estimators account for multiple observations (rows) for individuals, whereas the negative-binomial scale parameter accounts for potential overdispersion.[18] Models were adjusted for potential confounders listed following the relevant tables. Negative-binomial models adjusted for discrete intervals of time—such as single-year age intervals—are termed 'piecewise exponential models'[18,19] (similar results could be achieved using Cox regression, but discrete time models were chosen here for their pedagogical clarity). Subgroup-specific IRRs were estimated by assessing two-way interactions between vape status and each potential confounder using interaction terms.

Supplementary analyses

Previous studies on adolescent vaping and subsequent smoking initiation typically have not used time-to-event statistical methods with time-varying e-cigarette exposure measures, prompting us to conduct supplementary analyses to assess the consequences of such methods. For these analyses, we created a supplementary dataset in which the 'baseline'—akin to a baseline timepoint in a prospective cohort study—was defined as the start of the age-interval two before the participants' survey age interval. For instance, the baseline for a participant surveyed at 17 years of age was the start of their 15-year age interval. This means that participants can potentially be

Figure 1: Illustration of retrospective timelines of smoking/vaping initiation events across single-year age-intervals for 6 participants.



[a] Participant 2 initiated vaping prior to age 12, but is only followed from aged 12 (as are participants who never-vaped).

[b] Participant 6 contributes 0 person-years at risk of smoking initiation in the ever-vaped category because they initiated vaping after the outcome of smoking initiation.

Table 2: Time-at-risk and rates of smoking initiation by single-year age intervals.

Age interval (years)	Never-smoked at start of interval (n_1)	End of follow-up during interval (n_2)[b]	Person-years at risk of smoking initiation during interval (n_1-0.5n_2)[c]	Never-vaped [a]		Ever-vaped [a]		Unadjusted IRR for smoking initiation
				Initiated smoking/person-years at risk[c]	Rate[d] of smoking initiation	Initiated smoking/person-years at risk[c]	Rate[d] of smoking initiation	
12	5114	41	5093.5	34/5076	0.7	7/17.5	40.0	59.7
13	5073	76	5035	56/4945	1.1	20/90	22.2	19.6
14	4997	1063	4465.5	100/4233.75	2.4	59/231.75	25.5	10.8
15	3934	1314	3277	93/2937.75	3.2	72/339.25	21.2	6.7
16	2620	1323	1958.5	44/1606.25	2.7	52/352.25	14.8	5.4
17	1297	1297	648.5	15/471.75	3.2	25/176.75	14.1	4.4
Total:			**20478**	**342/19270.5**	**1.8**	**235/1207.5**	**19.5**	**11.0**

IRR = incidence rate ratio.
[a]Vape status is a time-varying covariate. Hence, participants who initiate vaping during an age interval will contribute half of their time-at-risk during the interval in the never-vaped category and half in the ever-vaped category.
[b]End of follow-up occurs during the age interval of smoking initiation for those who initiate smoking (12–17 years), or during the age interval of survey completion for those who never smoked (14–17 years).
[c]Participants who experience end of follow-up during an age interval contribute ½ person-year at risk during the interval (because end of follow-up is assumed to have occurred, on average, halfway through the interval). Participants who initiate vaping during an age interval have their time-at-risk equally divided between never-vaped and ever-vaped categories during the interval.
[d]Incidence rate per 100 person-years.

retrospectively followed-up for a maximum of 2.5 years (assuming once again that surveys occurred, on average, halfway through the survey age-interval). Participants who initiated smoking before baseline were excluded. The dataset was then analysed using three distinct negative-binomial regression methods: (1) the 'time-varying-exposure method' that updates vape status from its baseline value as a time-varying exposure variable where required while accounting for time-at-risk in the negative-binomial model (this is the same method used for the main analysis mentioned earlier); (2) the 'baseline-exposure method', which does not update vape status from its baseline value, but still accounts for time-at-risk; (3) the 'baseline-exposure relative risk (RR) method' that collapses the dataset to one row per participant, does not update vape status from its baseline value, and excludes time-at-risk from the negative-binomial model (meaning rate denominators became person-frequencies rather than person-years, and the effect measure became RR rather than IRR).

Sensitivity analyses

Sensitivity analyses assessed whether the association between vaping and subsequent smoking initiation was affected by participants with low lifetime smoking consumption. In these analyses, we re-ran the main analysis after excluding those who had only smoked <10 cigarettes and/or just a few puffs by the time of the survey.

Results

A total of 5243 adolescents aged 14–17 completed the wave 4 and/or 5 surveys; 756 completed both surveys, with only their later data included. Excluded were 13 adolescents who started smoking before the age of 12, 110 who were unclear about whether they initiated smoking or vaping first, and 7 with missing age at smoking initiation, leaving 5114 participants.

Eleven percent (n=577) of participants initiated smoking (Table 1), and 24% (n=1251) initiated vaping before end of follow-up. Both vaping and smoking prevalences were similar across most demographics but were higher for older adolescents and Aboriginal and Torres Strait Islanders than for non-indigenous adolescents in relation to smoking (26% vs 11%).

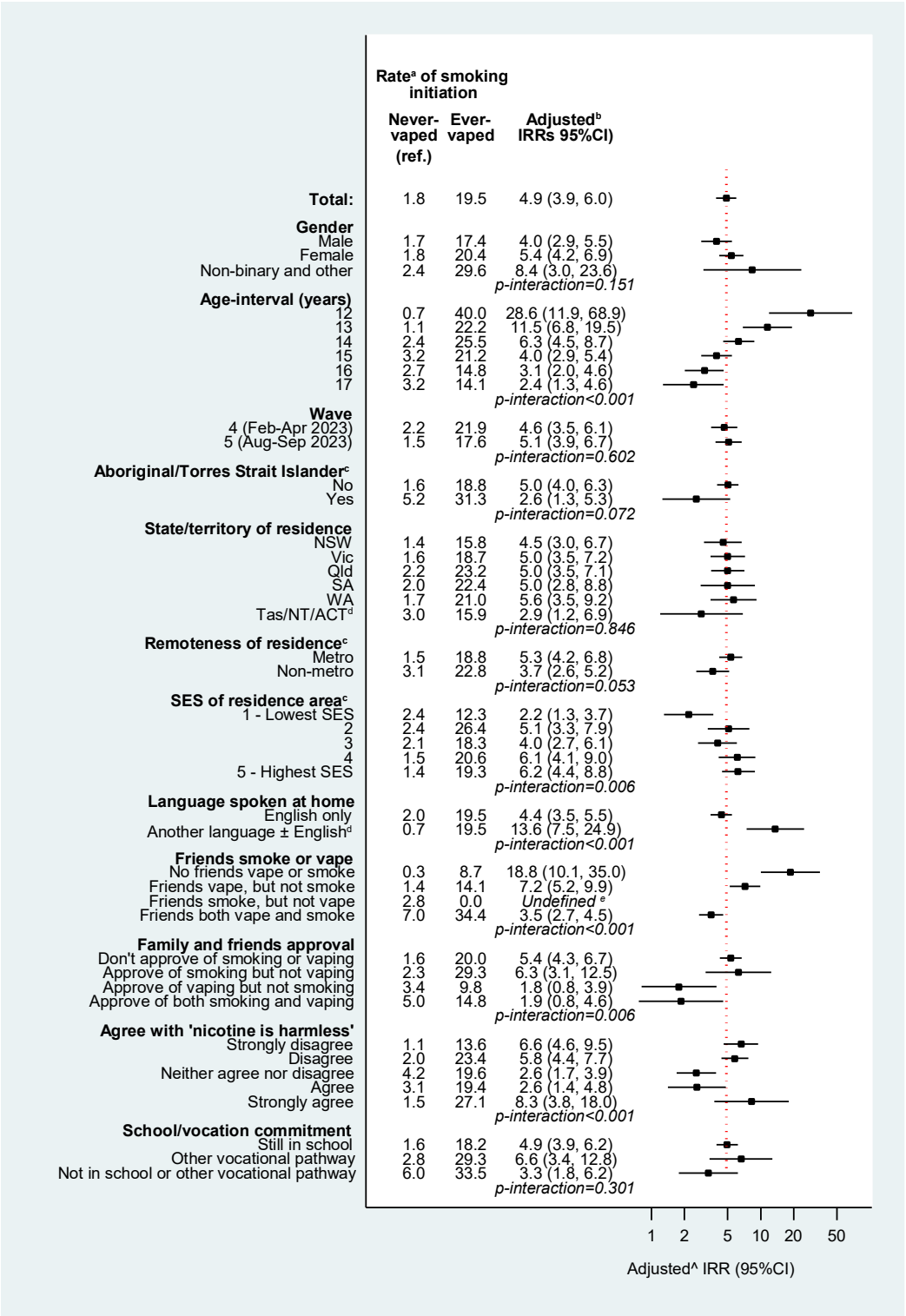
Participants were retrospectively followed for 20478 person-years (Table 2). Crude smoking initiation rates were 1.8 and 19.5 per 100 person-years for those who never-vaped and ever-vaped, respectively. After adjusting for all potential confounders, the

Table 3: Incidence rate ratios for smoking initiation according to the time-varying covariate vape status.

Vape-status	Initiated smoking/person-years at risk	Rate[a] of smoking initiation	Incidence rate ratios for smoking initiation			
			IRR_1 (95%CI)	IRR_2 (95%CI)	IRR_3 (95%CI)	IRR_4 (95%CI)
Total:	577/20478	2.8				
Never-vaped	342/19270.5	1.8	ref.	ref.	ref.	ref.
Ever-vaped	235/1207.5	19.5	11.0 (9.3, 12.9)	8.4 (6.9, 10.2)	8.2 (6.7, 10.0)	4.9 (3.9, 6.0)
p-value			<0.001	<0.001	<0.001	<0.001

CI = confidence interval; IRR = incidence rate ratio.
IRR_1 unadjusted.
IRR_2 adjusted for single-year age interval.
IRR_3 adjusted for same factors as IRR_2 plus wave, gender, Aboriginal or Torres Strait Islander status, state/territory of residence, remoteness of residence, socioeconomic status of residence area, and language spoken at home.
IRR_4 adjusted for same factors as IRR_3 plus whether friends smoke and/or vape, family/friends views on smoking/vaping, perceived harmlessness of nicotine, and school/vocation commitment.
[a]Incidence rate per 100 person-years.

Figure 2: Adjusted incidence rate ratios (IRRs) for smoking initiation according to vape-status by subgroup.



IRR = incidence rate ratio.

[a] Incidence rate per 100 person-years.

[b] IRRs adjusted for single-year age interval, wave, gender, Aboriginal/Torres Strait Islander status, state/territory of residence, remoteness of residence, socioeconomic status (SES) of residence area, language spoken at home, whether friends smoke and/or vape, family/friends views on smoking/vaping, perceived harmlessness of nicotine, and school/vocation commitment.

[c] "Not known" and "prefer not to answer" categories excluded.

[d] Categories combined due to low numbers.

[e] IRR undefined due to rate = 0% in ever-vaped category.

estimated smoking initiation rate was nearly 5 times higher for those who ever-vaped than for those who never-vaped (IRR=4.9; 95% confidence interval {CI}: [3.9, 6.0], $p<0.001$)(Table 3). The association between vaping and subsequent smoking significantly varied across demographic subgroups (Figure 2). Notably, IRRs were considerably higher at younger ages, with 12-, 13-, and 14-year-olds having

estimated IRRs of 28.6 (95%CI: [11.9, 68.9]), 11.5 (95%CI: [6.8, 19.5]), and 6.3 (95% CI: [4.5, 8.7]), respectively, than an IRR of 2.4 (95% CI: [1.3, 4.6]) for 17-year-olds (p-interaction<0.001). For SES of residence area, the highest SES group had an IRR of 6.2 (95% CI: [4.4, 8.8]), significantly higher than the IRR of 2.2 (95% CI: [1.3, 3.7]) for the lowest SES group (p-interaction=0.006). Those speaking another language in addition to English had an IRR of 13.6 (95% CI: [7.5, 24.9]), considerably higher than the IRR of 4.4 (95% CI: [3.5, 5.5]) for English-only speakers (p-interaction<0.001).

Applying the 'time-varying-exposure method' to the supplementary dataset yielded an adjusted IRR of 4.1 (95% CI: [3.3, 5.1]) (eTable 1). In contrast, the 'baseline-exposure method'—which did not update baseline vape-status as a time-varying covariate—estimated a lower IRR of 2.3 (95% CI: [1.5, 3.5]), thus diluting the estimated IRR by 44% (i.e. from 4.1 to 2.3). This regression dilution primarily stems from misclassification in that 42% (n=178) of the 425 participants who had never-vaped at baseline but later initiated smoking had actually initiated vaping between baseline and smoking initiation. For these participants, the 'baseline-exposure method' incorrectly attributed their smoking initiation to the never-vaped category. The 'baseline-exposure RR method'—which also incorrectly attributed 178 adolescents who initiated smoking to the never-vaped category, while additionally assuming that time at risk was equal for all participants—produced an RR of 1.9 (95% CI: [1.3, 2.7]).

The sensitivity analyses showed that excluding those who initiated smoking but had smoked <10 cigarettes did not change the IRR estimate of 4.9 (eTable 2).

Discussion

Our study is the first to estimate the relationship between vaping and subsequent smoking initiation in an Australian cohort of adolescents. Through analysis of Australian adolescents retrospectively followed from age 12 to 17 years, we found that the rate of smoking initiation was nearly 5 times higher among those who had previously initiated vaping than among those who had not. Our findings not only reveal the high risk of smoking initiation that vaping imposes on Australian adolescents but also contribute to the broader global understanding of vaping's role in smoking initiation.

In a 2021 systematic review, a meta-analysis of 17 prospective studies across 9 countries found that among adolescents less than 20 years of age, those who had ever-vaped at baseline were at three times the risk of subsequent smoking initiation than those who had never-vaped at baseline (pooled adjusted RR=3.0 [95% CI: {2.4, 3.8}]).[10] However, when we critically assessed these 17 studies, we found a recurring methodological limitation: they did not account for the time-varying nature of e-cigarette exposure. In our supplementary analyses, when estimating 'relative risk' using analytical methods analogous to previous studies (the 'baseline-exposure RR method'), our RR estimate of 1.9 was lower in magnitude than the pooled RR estimate of 3.0 reported in the systematic review. Furthermore, when e-cigarette exposure was not updated from its baseline value, the IRR estimate from the 'baseline-exposure method' was diluted by 44% compared to the IRR estimate derived using the 'time-varying-exposure method' (from 4.1 to 2.3). Although IRRs and RRs have different interpretations and are not directly comparable, the 44% dilution observed in our IRR estimate (due entirely to not accounting for time-varying e-cigarette exposure) implies that RR estimates—and by natural extension, odds ratio estimates—in previous studies from other regions may be understated.

It is unclear why the prospective cohort studies in the systematic review did not account for time-varying measures of e-cigarette exposure. One reason may be the inherent design of prospective cohort follow-up questionnaires, which typically assess participants' status at a specific timepoint rather than retrospectively eliciting details of changes between baseline and follow-up. The retrospective nature of our study compelled us to consider this issue, leading us to add questions that allowed us to construct timelines of initiation events. If questionnaires do not specifically ask about the sequence and timing of smoking and vaping initiations between baseline and follow-up, analysts will be constrained to using baseline vape status as a time-invariant measure. While such an approach can be appropriate if the underlying values of the exposure of interest are relatively stable over time, our supplementary analysis demonstrates the significant regression dilution when adolescent vape-status is not updated. Additionally, while we have highlighted the pitfalls of relying solely on the baseline value of vape status without subsequent updates, researchers should also be cognisant of the risks associated with designating vape status as time-invariant using its end-of-follow-up value only. This method introduces the potential for an even more insidious problem known as 'immortal time bias'.[20] While an exploration of immortal time bias is beyond the scope of this paper, its risks—like regression dilution—can be mitigated by properly accounting for time-varying exposures.[20]

As previously noted, to our knowledge, only one prior study has accounted for time-varying e-cigarette exposure when assessing subsequent smoking initiation risk among adolescents.[11] In that study, treating e-cigarette use as a time-varying exposure indicated that among 9584 e-cigarette- and cigarette-naïve US adolescents aged 12 to 17 years at baseline, e-cigarette initiators were 2.4 times more likely to subsequently start cigarette smoking than non-e-cigarette initiators. However, because the study did not simultaneously analyse e-cigarette initiation as a time-invariant exposure, it remains unknown how much regression dilution the time-varying method averted.

Another unique aspect of our study is the comparison of age-specific effect estimates (Figure 2), a feature absent in the 17 studies included in the systematic review. By using time-to-event methods, we were able to divide time-at-risk of smoking initiation into single-year age intervals, thereby allowing individuals to contribute data across multiple age intervals and thus retain relatively high precision in our age-specific estimates. The age-specific results revealed an alarming trend: the younger an individual starts vaping, the higher their relative rate of smoking initiation. For example, 12-, 13-, and 14-year-olds had estimated IRRs of 28.6, 11.5, and 6.3 for smoking initiation, respectively, compared to an IRR of 2.4 for 17-year-olds. These figures highlight a stark gradient of relative risk, with younger adolescents incurring the most severe consequences of a poorly regulated e-cigarette market. While e-cigarettes may have been initially claimed as a smoking cessation aid for adult smokers, the stark disparities in IRRs for younger adolescents are a sobering reminder of the broader public health consequences of a market that has failed to contain vaping product marketing and access.

Our sensitivity analyses demonstrated that excluding adolescents who might only be considered to have experimentally smoked (i.e. fewer than 10 cigarettes) did not alter the IRR estimate of 4.9

(eTable 2). In other words, the initiation of vaping appears as likely to lead to smoking beyond non-experimental levels as to mere initiation. Consistent with this notion are additional results from the systematic review mentioned earlier,[10] which found that the pooled RR of 3.0 (for ever-smoking) was only marginally reduced to 2.6 when the outcome was cigarette use in the past 30 days, frequent, and/or daily use. These findings align with broader concerns about the escalatory nature of tobacco use among adolescents, where even small amounts of initial cigarette smoking have been shown to predict subsequent escalation to regular use,[21] which in turn considerably increases the risk of addictive smoking during adulthood.[22]

Australia has observed a general declining trend of smoking uptake among 12- to 17-year-old adolescents since about the mid-1990s,[7] but a recent analysis of bi-annual tracking data from 2018 to 2022 suggests a possible increase in current smoking among 14- to 17-year-olds, from 2.1% to 6.7%, alongside a rise in current vaping from 0.8% to 11.8%.[23] While these concurrent increases in both smoking and vaping rates naturally prompt questions about a potential link, a critical limitation of the tracking study is its lack of data on smoking prior to the widespread use of e-cigarettes in Australia. Another data source, the triennial ASSAD survey,[4] provides insights going back to 1984 and is perhaps suggestive of a marginal slowing in the decline of adolescent ever-smoking rates after e-cigarettes were introduced around 2010/11; however, more empirical data from reliable sources are needed to fully understand how e-cigarettes might be influencing adolescent smoking rates at the population-level.

Our study has limitations. First, because our survey asks for age at key events in whole years, approximations of time-at-risk are subject to rounding error. For example, a participant starting smoking at the age of 15 yields a 3.5-year time-at-risk from age 12, but this could vary between 3 and almost 4 years, depending on their exact initiation date. Such rounding errors, however, are non-differential by vape-status, thus imposing a low risk of bias. Second, although we adjusted for a comprehensive set of seven socio-demographic and four common liability confounders—similar to various confounder sets used in multiple studies judged to be appropriately adjusted and/or at low-risk of confounding bias in multiple systematic reviews[10,24–27]—there is always a chance of an important unmeasured confounder in any observational study. Moreover, our retrospective design means that confounder values reflect participants' circumstances at the time of survey and not necessarily prior to vaping or smoking initiation. Nonetheless, our 'baseline-exposure RR' result suggests no marked upwards bias arising from unmeasured or imperfectly measured confounders, given that our adjusted RR estimate of 1.9 was lower than the pooled estimate of 3.0 reported in the systematic review.[10] Fourth, a potential source of selection bias in all retrospective cohort analyses is non-ignorable, exiting[15] which could occur in our analysis if potential participants' vaping or smoking status was independently associated with their availability for study recruitment. However, given that the ever-smoking and ever-vaping rates in our cohort are very close to those reported in the most recent ASSAD survey,[4] we have little reason to suspect that our analysis was significantly vulnerable to this type of selection bias.

The debate around the population health effects of e-cigarettes typically revolves around whether the potential benefits of vaping—such as assisting some people who smoke to quit—outweigh the risks.[28] For this to be an informed debate, however, the risks and benefits need to be accurately quantified. Our results suggest the increased risk of smoking uptake after vaping initiation is high for Australian adolescents, with those aged 12, 13, and 14 bearing an alarmingly disproportionate burden of the risk. This significant age-related disparity underscores the need for public health policies aligned with Australia's National Tobacco Strategy, notably in preventing e-cigarette uptake among young people and non-smokers.[29] In a broader context, our findings suggest potential underestimation of the increased risks of smoking due to vaping in studies of adolescents from other regions of the world. To address this, future studies should design their questionnaires to enable the use of time-to-event methods, and then actively employ these methods, thus ensuring a more accurate assessment of vaping risks in adolescent populations.

Ethics approval

Ethical approval for the study was provided by The University of Sydney Human Research Ethics Committee. Project number: 2021/442.

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Conflicts of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A Supplementary data

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